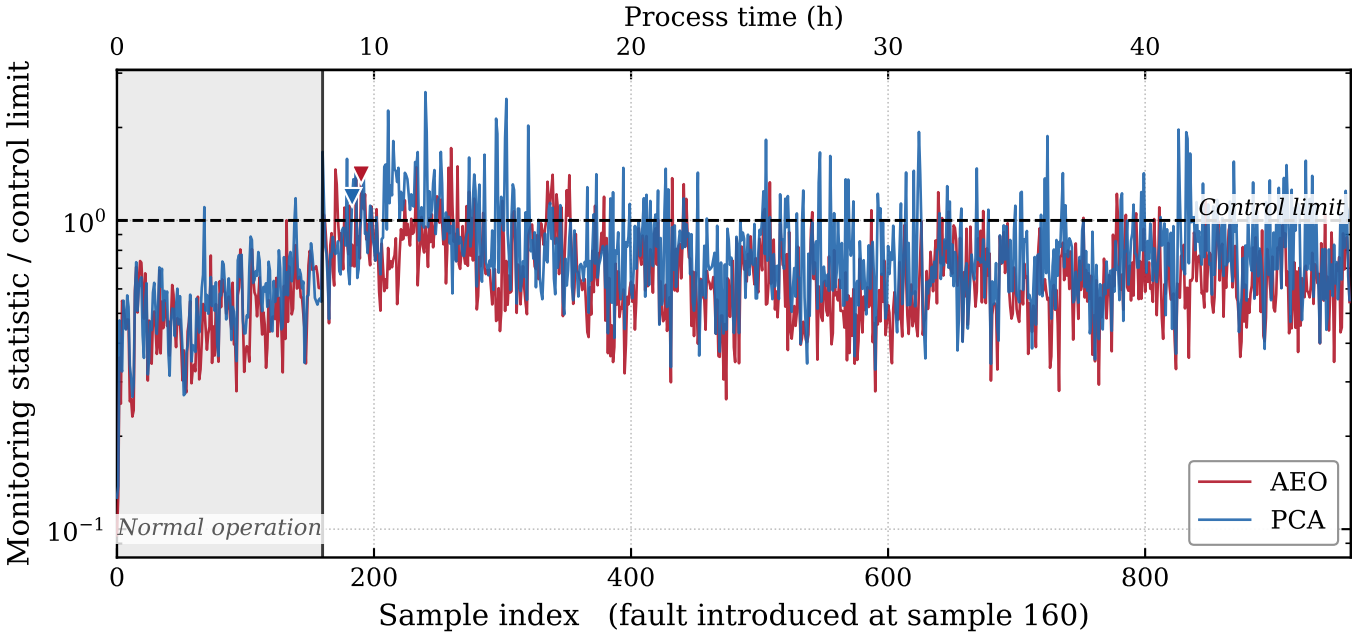
Fault Detection Performance for IDV(19)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	299.3 samples (898 min)	MDR 0.939	detected in 6/7 runs
PCA	delay	78.1 samples (234 min)	MDR 0.768	detected in 7/7 runs

PCA detects 221.2 samples (664 min) earlier; PCA missed-detection rate lower by 18%